Spatial Triple-Correlation Spectroscopy (S3CS) Reveals Heterotrimer Dynamics in Live Cells

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The provided code for spatial triple correlation microscopy is designed for researchers seeking to quantify interaction among protein subunits within individual complexes and to assess associations between distinct protein assemblies in live cells. The code implements high-order correlation analysis to detect the formation and dynamics of heterotrimeric protein complexes and to distinguish between reversible and stable interaction processes across subcellular compartments. The intent is to enable the spatial mapping of heterotrimer mobility within intracellular environments and allow for the accurate identification of fully assembled heterotrimeric complexes relative to substoichiometric intermediates. Improvements and open-source data will be continuously included to facilitate accessibility and adoption of the method.

Files required

To start using the code for pair correlation microscopy, the "bfmatlab" folder, which is required to open TIFF files, needs to be downloaded. "bfmatlab.zip" can be downloaded from "https://www.openmicroscopy.org/bio-formats/downloads/". After downloading "bfmatlab.zip", unzip the folder and save it to the folder containing all MATLAB functions for spatial triple correlation microscopy.

Getting Started

S3CS workflow for three-channel confocal line scan data

Open the file called "*S3CS_Workflow_LineScans*" and follow the steps indicated within the code. Details are provided below:

1. Run Step 1 to define the folder with all MATLAB functions.
2. Run Step 2 to open TIFF files. In this section, select a single or multiple TIFF files.
   NOTE: Each TIFF file from the LSM880 microscope contains data for the three channels, the code will assign each channel to Data1, Data2, and Data3 cell arrays, and then it will remove the first 10 000 lines from each line scan.
3. Run Step 3 to fragment the data from each channel. The size of the fragments is indicated in *'fragmentSize'*. A fragment size of 20 000 lines corresponding to ~7 s acquisition time is recommended.
4. Run Step 4 to calculate the ACF for each fragment to eliminate artifacts from long-term instabilities or single bright events. Change the value in *'SampFreq'* with the correct sampling frequency.

5. Run Step 5 to establish the threshold for the standard deviation to systematically discard fragments with artifacts. For example, if *'mumberSD'* = 3 is indicated, any segment whose ACF exhibited a standard deviation exceeding three times the mean standard deviation will be discarded.
   NOTE: Evaluate the 'Before' and 'After' ACF plots to find the correct threshold (Fig. 1).

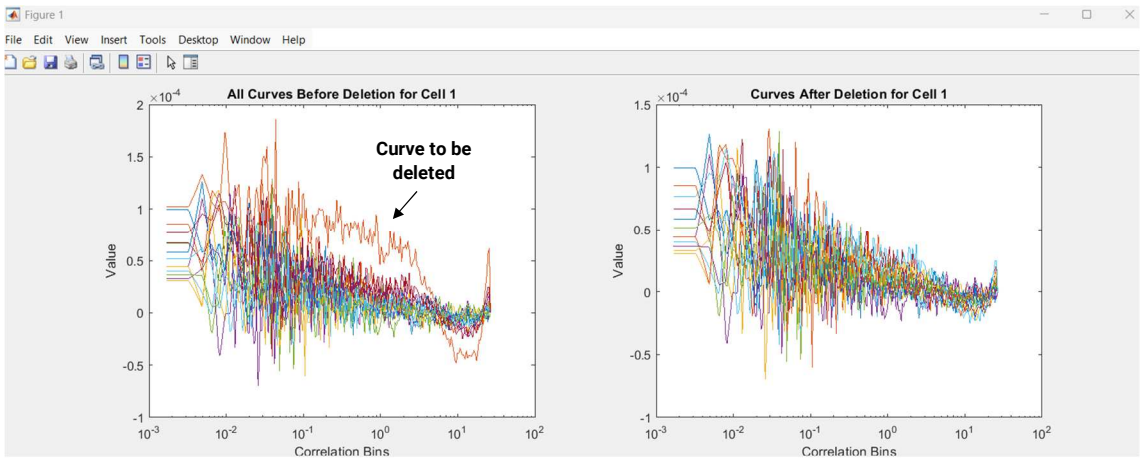

Figure 1. ACF plots before and after curve exclusion. On the right, all curves produced after the fragmentation of the time series are shown. On the right, after systematically discarding segments with artifacts, only the curves to be included in the final analysis are shown.

6. Run Step 6 to delete the fragments with strong photobleaching or cell movement identified in Step 5.
   NOTE: If a fragment with strong photobleaching or cell movement is identified in one channel but not in the other two, the function will automatically delete that segment from the three channels to keep the temporal information.
7. Run Step 7 to order the data.
8. Run Step 8. Get the average matrix to perform S3CS analysis. After running this step, a message will appear in the command window. Answer *'yes'* to round the average values or answer *'no'* to consider decimal points in the average data (if no answer is provided, *'no'* is considered as the default answer).
9. **OPTIONAL:** Reverse the order of the average matrix ('*Fragmented_Data1_avg*') for bidirectional analysis. If this function is selected, the first column from each line scan will become the last column, and the last column will become the first column.
10. **Step 9. Calculate the triple correlation.** In this step, enter the values to be considered to perform triple-autocorrelation or spatial-triple correlation analysis. The variables to be considered are: *i)* distance between pixels ('Dist'), if '*Dist* = 0', a triple-autocorrelation will be calculated, if the distance is greater than zero, e.g., '*Dist* = 3', the spatial-triple correlation will be calculated; *ii)* sampling frequency (*'SampFreq'*); *iii)* binsize (*'m'*): the larger the *m* value, the higher the computational cost; *iv)* first and last column to be analyzed (*'FirstCol'* and *'LastCol'*). For this last parameter, the first and last columns will be applied to all line scans.

NOTE: If bidirectional analysis is performed, activate the lines that use reverse data as input (Fig. 2). Be sure to reverse the order of the average matrix before performing bidirectional analysis by running the optional section indicated above.

Triple correlation function analysis **using the direction of the line scan**

```
%% Step 9. CALCULATE TRIPLE CORRELATION

Dist     = 0;    % 'Dist' = 0 for triple correlation analysis. 'Dist' > 0 to calculate cross-triple correlation
SampFreq = 3415; % 3415 for 300,000 lines using 7.81 us pixel dwell time. Change sampling frequency as needed.
m        = 220;  % BinSize for multiple tau approach. Recommended value.
FirstCol = 1;    % First column to be analyzed
LastCol  = 16;   % Last column to be analyzed

[BspecCellArray, diagonalElementsAbs,corrBins, corrResult_array, F3CS_avg, avg_BspecCellArray,tau1,tau2] = plot3FCS_MulTau3_v3...
    (Fragmented_Data1_avg, Fragmented_Data2_avg, Fragmented_Data3_avg, FirstCol, LastCol, Dist, m, SampFreq);

% Activate the following lines if you want to perform bidirectional analysis (reverse orientation)

%[BspecCellArray, diagonalElementsAbs,corrBins, corrResult_array, F3CS_avg, avg_BspecCellArray,tau1,tau2] = plot3FCS_MulTau3_v2...
%    (Rev_Fragmented_Data1, Rev_Fragmented_Data2, Rev_Fragmented_Data3, FirstCol, LastCol, Dist, m, SampFreq);
```

Triple correlation function analysis **reversing the order of the line scan**

```
%% Step 9. CALCULATE TRIPLE CORRELATION

Dist     = 0;    % 'Dist' = 0 for triple correlation analysis. 'Dist' > 0 to calculate cross-triple correlation
SampFreq = 3415; % 3415 for 300,000 lines using 7.81 us pixel dwell time. Change sampling frequency as needed.
m        = 220;  % BinSize for multiple tau approach. Recommended value.
FirstCol = 1;    % First column to be analyzed
LastCol  = 16;   % Last column to be analyzed

%[BspecCellArray, diagonalElementsAbs,corrBins, corrResult_array, F3CS_avg, avg_BspecCellArray,tau1,tau2] = plot3FCS_MulTau3_v3...
%    (Fragmented_Data1_avg, Fragmented_Data2_avg, Fragmented_Data3_avg, FirstCol, LastCol, Dist, m, SampFreq);

% Activate the following lines if you want to perform bidirectional analysis (reverse orientation)

[BspecCellArray, diagonalElementsAbs,corrBins, corrResult_array, F3CS_avg, avg_BspecCellArray,tau1,tau2] = plot3FCS_MulTau3_v2...
    (Rev_Fragmented_Data1, Rev_Fragmented_Data2, Rev_Fragmented_Data3, FirstCol, LastCol, Dist, m, SampFreq);
```

Figure 2. Reversing the order of the line scan for bidirectional analysis. On the top panel, parameters to perform the triple correlation function using the direction of the line scan. On the bottom panel, parameters to perform the triple correlation function reversing the order of the line scan. Here, '*Rev_Fragmented_Data1, 2, and 3*' include data where the last column of each line scan became the first column and the first column became the last column.

11. **Step 10.** Plot results in a carpet representation. In this step, the values along the $\tau_1 = \tau_2$ line from each correlation matrix are extracted and visualized in a carpet representation. Each column in the correlation carpets corresponds to a point along the detection line where S3CS was calculated, with the y-axis representing correlation time on a logarithmic scale (Fig. 3).

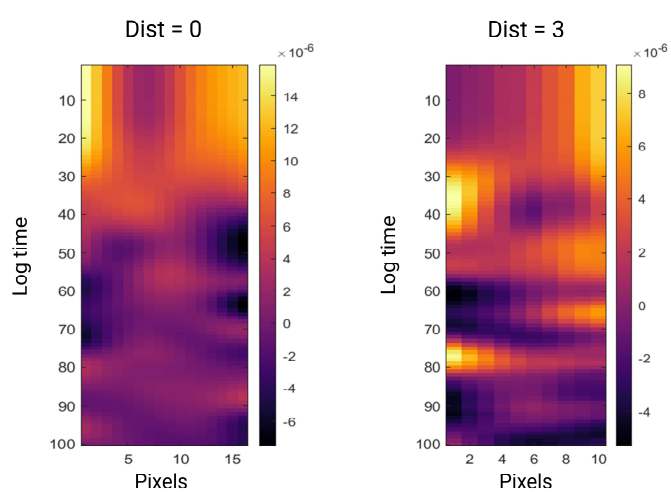

Figure 3. Carpet representation for triple correlation data. On the right, carpet representation of the triple-autocorrelation function. On the left, carpet representation of the spatial-triple correlation function. The values along the $\tau_1 = \tau_2$ line from each correlation matrix for every pixel along the line scan are visualized. In each carpet, columns correspond to a point along the detection line where the triple correlation was calculated, with the y-axis representing correlation time on a logarithmic scale.

12. **Step 11.** Plot the average 3D surface for all τ_1 and τ_2 delay times. This function plots the average S3CS profile from all pixels in the line scan (Fig. 4).

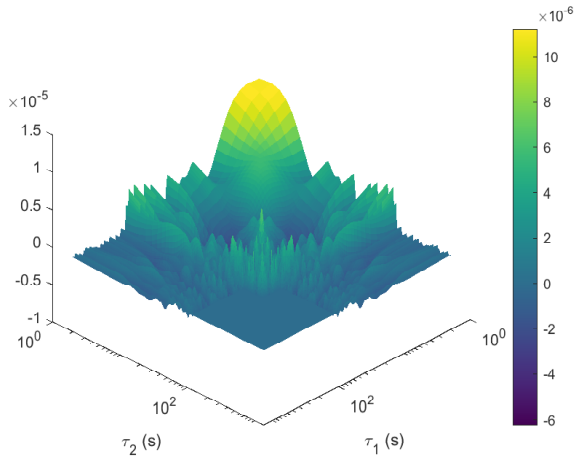

Figure 4. Triple correlation profile. Average S3CS profile for all τ_1 and τ_2 delay times.

13. **OPTIONAL.** Calculate and plot principal component (PC) analysis to distinguish data from heterotrimeric protein complexes and independent proteins (for visualization purposes only) (Fig. 5). This function uses as input the output from S3CS analysis.

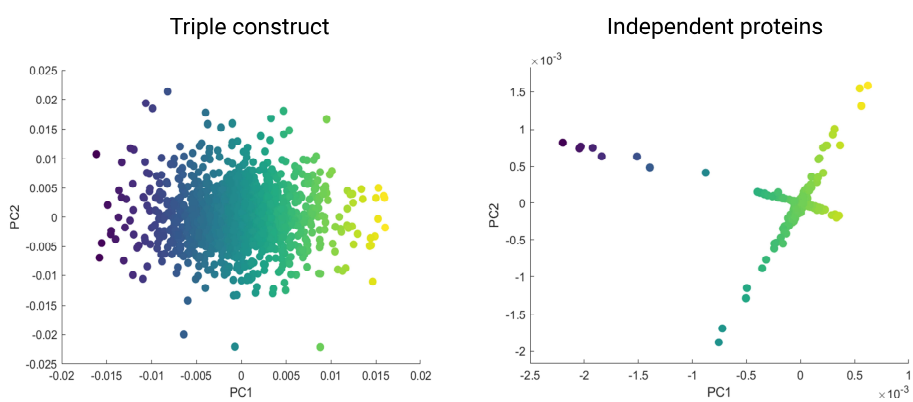

Figure 5. Principal component (PC) analysis using S3CS results. On the left, PC analysis using TCCS results from a cell transfected with a triple construct. On the right, PC analysis using S3CS results from a cell transfected with independent proteins.

14. **Step12.** This step calculates the fraction of triple correlation from the three fluorescence channels. The function first computes the single-channel autocorrelation functions for the first, second, and third channels. It then identifies the limiting channel, defined as the channel exhibiting the highest complex concentration. The triple-correlation fraction is obtained by normalizing the maximum amplitude of the S3CS function (G_{123}) to the amplitudes of the corresponding single-channel autocorrelation functions, therefore accounting for the channel containing the limiting number of molecules. In this step, you should modify only the following parameters: *'firstCol'*, first column to be analyzed; '*lastCol'*, last column to be analyzed; and *'sampleFreq'*, acquisition sampling frequency. A correction factor ('*correction_factor*') must be determined independently

for each microscope to account for γ_3 and γ_2, which are correction factors accounting for the effective observation volumes associated with triple and double correlation measurements.

15. Run **Step 13** to fit the spatial-S3CS to a Gaussian for $\delta r > 0$ (when '*Dist*' was set to a value different that 0 in Step 9). This subsection will fit the spatial-triple correlation result along the $\tau_1 = \tau_2$ line to a Gaussian model using the *fit* function in MATLAB. Consider that if two molecular species differ in diffusion coefficient by more than a decade, they will appear as separate peaks in the correlation profile. Hence, indicate if the fit will be performed using either one or two-term Gaussian models by answering '*gauss1*' or '*gauss2*' in the command window. If no answer is provided, the code will fit the spatial-S3CS profile to a one-term Gaussian model. We recommend using one Gaussian model, visualizing the fitting, and deciding if the data requires a two-term Gaussian model.
16. Run Step **14** to plot the raw and fitted S3CS data, and get the peak values ('*nPeakTable*'). The peak values obtained correspond to the estimated translocation times of the complex.

NOTE: If fitting the S3CS to a Gaussian using default starting values and constraint bounds results in inaccurate results (i.e., the peak is not determined accurately), use optimized starting values using the MATLAB Curve Fitting Toolbox as indicated in the code (Steps 15-18). To open the MATLAB Curve Fitting Toolbox, write *"curveFitter"* in the command window. A complete user guide can be found at *https://au.mathworks.com/help/curvefit/*. Note that when entering the data to the Curve Fitting toolbox, the time corresponding to the *x*-axis should be on a logarithmic scale; perform this calculation before inter the x-axis to the toolbox. For example, if '*logTime = log(corrBins)*', then '*logTime*' is the dataset to enter as the x-axis.

17. In **Step 15**, indicate which correlation profiles will be fitted using optimized starting values, then run this step.
18. In **Step 16**, replace the values in *''opts0.StartPoint"* with the values obtained from the MATLAB Curve Fitting Toolbox. Replace the values for each dataset (values for *pCF_Cell1, 2, 3,* and so on). Run this section to get the new fitted values.
19. In **Steps 17** and **18**, indicate the number of fitted curves we want to obtain. For example, if we fitted three pCFs using optimized starting values, then change the number for *"numDatasets ="* to "numDatasets = 3". Run these sections to obtain the new fitted data ('*Adit_fvalues_pcf*').